Market Intelligence case: Company S vs Company X

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Editable files (for the live session)
GitHub repo (SQL, results, mock database, Q4 note): `https://github.com/hanh-le-hong/market-intelligence-case`
Q5 model (Google Sheets, 9 tabs): `https://docs.google.com/spreadsheets/d/1LgIXNfAwEwsVd1FwZXyYwN49rmuzRnWOJmo0zYuTAfQ`
Q4 note (Google Docs): `https://docs.google.com/document/d/1weGjmNjq-L1TKyGHXxeGyTWV5MOia0XOcUD4bwYUfFU`

How the SQL part was done

No data was provided, so I built a mock PostgreSQL database that follows the schema and Appendix A exactly (10 tables, 100 S shops, 100 X shops, 60 dual pairs, 2,500 products, 37,479 orders, months 2026-07 to 2026-10), then audited it against Appendix A (18 checks, file `sql/00b_audit_vs_appendix.sql`) before writing the queries. The mock data contain deliberate edge cases (a shop that becomes dual mid-period, a missing crawl month, shop-months where crawler and market research disagree) so the queries are tested against them. All queries run as-is on that database; the result files in `results/` are their output. Dialect: PostgreSQL. T-SQL differences are noted in the files.

Assumptions shared by Q1 to Q3

- **Last 3 months** = the three latest months in the data: 2026-08, 2026-09, 2026-10. Hard-coded for readability; a dynamic version is noted in the files.
- **Sold** = `order_status = 'complete'`. Created / paid / on_delivery are not final; canceled / returned are not sales. X crawler GMV also cannot contain canceled orders, so this keeps S and X comparable.
- **GMV** as defined in Appendix A: before discount and shipping deduction (`item_gmv`, `gmv`). All values in USD.
- **Days** for ADG / ADO = calendar days of the month (31 / 30 / 31), not days with sales. **ABS** = GMV / orders. Tier bounds: lower inclusive, upper exclusive.
- Top-N uses `ROW_NUMBER()`, so exactly N rows per month; ties at the cut-off are broken arbitrarily.

A1. Top 100 selling items each month (by GMV), company S

Business question: which products carry S's GMV each month, at what price, and how many orders.

Specific assumptions. Item = `s_product_id`; details from `dim_s_product` (name, brand, category lv1/lv2). Average selling price = quantity-weighted average of `discounted_price` (what the buyer paid), not the listing price; min / max = min / max of `discounted_price`. order_count = `COUNT(DISTINCT order_id)` because a product can appear on more than one item row of the same order.

Template (one row = one product in one month; 300 rows). First rows of the actual result:

order_month	rank	s_product_id	brand	category_lv1	avg_price	min	max	gmv	orders
2026-08	1	fc89c75a-…	NoBrand	Home Appliances	450.00	360	468	2,808	4
2026-08	2	f8be6b87-…	NoBrand	Home Appliances	349.50	345	354	2,262	2
2026-08	3	63c8c029-…	NoBrand	Home Appliances	215.57	196	258	1,806	5

Columns in the file: order_month, rank_in_month, s_product_id, product_name, brand, category_lv1, category_lv2, avg_price, min_price, max_price, gmv, order_count. Full result: `results/q1_result.csv`.

Query (`sql/02_q1_top100_items.sql`):

```
WITH item_month AS (
    SELECT i.order_month, i.s_product_id, p.product_name, p.brand, p.category_lv1, p.category_lv2,
           SUM(i.item_gmv)                                                          AS gmv,
           COUNT(DISTINCT i.order_id)                                               AS order_count,
           ROUND(SUM(i.discounted_price * i.quantity)::numeric / SUM(i.quantity), 2) AS avg_price,
           MIN(i.discounted_price)                                                  AS min_price,
           MAX(i.discounted_price)                                                  AS max_price
    FROM fact_s_order_item AS i
    JOIN fact_s_order   AS o ON o.order_id     = i.order_id
    JOIN dim_s_product  AS p ON p.s_product_id = i.s_product_id
    WHERE o.order_status = 'complete'
      AND i.order_month IN ('2026-08', '2026-09', '2026-10')
    GROUP BY i.order_month, i.s_product_id, p.product_name, p.brand, p.category_lv1, p.category_lv2
),
ranked AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY order_month ORDER BY gmv DESC) AS rank_in_month
    FROM item_month
)
SELECT order_month, rank_in_month, s_product_id, product_name, brand, category_lv1, category_lv2,
       avg_price, min_price, max_price, gmv, order_count
FROM ranked
WHERE rank_in_month <= 100
ORDER BY order_month, rank_in_month;
```

Why two CTEs: a window function cannot sit in WHERE, so the rank is computed first and filtered in the next step. T-SQL: identical; replace `::numeric` with `CAST(... AS decimal)`.

A2. Top 20 dual shops in S each month, with ADG / ADO / ABS / ABS tier

Business question: among sellers already on both platforms, who are S's biggest 20 each month and what does their basket look like.

Specific assumptions. Dual = the S shop has a row in `fact_s_x_monthly_shop_mapping` for *that month* (the mapping is monthly, so dual status is evaluated per month: a shop can be dual in Sep and not in Aug). GMV and order count come from `fact_s_order` (one row per order). `x_shop_id` is carried from the mapping so Q3 can join X data directly.

Template (one row = one dual S shop in one month; 60 rows). Sample rows:

order_month	rank	shop_name	seller_type	days	gmv	orders	adg	ado	abs	abs_tier
2026-08	1	Nam Zone 54	Non-mall	31	19,975	58	644.35	1.87	344.40	over 50 USD
2026-08	2	Tefal Official Store	Mall	31	13,451	58	433.90	1.87	231.91	over 50 USD
2026-08	5	men_tuấn547	Non-mall	31	8,437	290	272.16	9.35	29.09	10 - 50 USD
2026-09	11	Walfos Official Store	Mall	30	5,339	280	177.97	9.33	19.07	10 - 50 USD

Columns in the file: order_month, rank_in_month, s_shop_id, shop_name, seller_type, x_shop_id, days_in_month, gmv, order_count, adg, ado, abs, abs_tier. Full result: `results/q2_result.csv`.

Query (`sql/03_q2_top20_dual_shops.sql`):

```
WITH shop_month AS (                      -- S performance per (shop, month), dual shops only
    SELECT o.order_month, o.s_shop_id, m.x_shop_id,
           SUM(o.gmv) AS gmv, COUNT(*) AS order_count
    FROM fact_s_order AS o
    JOIN fact_s_x_monthly_shop_mapping AS m
      ON  m.s_shop_id     = o.s_shop_id
      AND m.mapping_month = o.order_month   -- dual status is per month
    WHERE o.order_status = 'complete'
      AND o.order_month IN ('2026-08', '2026-09', '2026-10')
    GROUP BY o.order_month, o.s_shop_id, m.x_shop_id
),
metrics AS (
    SELECT sm.*,
           EXTRACT(DAY FROM (TO_DATE(sm.order_month || '-01', 'YYYY-MM-DD')
                             + INTERVAL '1 month' - INTERVAL '1 day'))::int AS days_in_month,
           ROUND(sm.gmv::numeric / sm.order_count, 2)                      AS abs
    FROM shop_month AS sm
),
ranked AS (
    SELECT mt.*, ROW_NUMBER() OVER (PARTITION BY mt.order_month ORDER BY mt.gmv DESC) AS rank_in_month
    FROM metrics AS mt
)
SELECT r.order_month, r.rank_in_month, r.s_shop_id, s.shop_name, s.seller_type, r.x_shop_id,
       r.days_in_month, r.gmv, r.order_count,
       ROUND(r.gmv::numeric         / r.days_in_month, 2) AS adg,
       ROUND(r.order_count::numeric / r.days_in_month, 2) AS ado,
       r.abs,
       CASE WHEN r.abs < 5  THEN 'under 5 USD'
            WHEN r.abs < 10 THEN '5 - 10 USD'
            WHEN r.abs < 50 THEN '10 - 50 USD'
            ELSE                 'over 50 USD' END AS abs_tier
FROM ranked AS r
JOIN dim_s_shop AS s ON s.s_shop_id = r.s_shop_id
WHERE r.rank_in_month <= 20
ORDER BY r.order_month, r.rank_in_month;
```

T-SQL: days_in_month = `DAY(EOMONTH(CAST(order_month + '-01' AS date)))`. Checks run: 20 rows per month; days = 31/30/31; abs × orders ≈ gmv; a shop that is dual only from September does not appear in August.

A3. Extend A2 with X shop performance (crawled_fact_x_shop)

Business question: for S's top 20 dual sellers, how does the same seller perform on X in the same month, side by side.

Specific assumptions. X row matched on `x_shop_id` (from the monthly mapping) and `crawled_month = order_month`. **LEFT JOIN**: a top-20 S shop stays in the list even when the crawler has no row that month; X metrics are NULL and tier = 'no X data'. Dropping it would silently change the top 20 and hide a data gap. X metrics are crawler estimates: gaps under ~10-15% vs S should not be read as real differences. Same formulas and tier thresholds on both sides. `gmv_ratio_x_vs_s` added as a convenience (> 1 = seller sells more on X).

Template (one row = one dual S shop in one month; 60 rows, same keys as A2):

month	rank	S shop	X shop	s_gmv	s_adg	s_ado	s_abs	s_tier	x_gmv	x_adg	x_ado	x_abs	x_tier	X/S
2026-08	1	Nam Zone 54	MaiShop.vn	19,975	644.35	1.87	344.40	over 50	17,100	551.61	1.26	438.46	over 50	0.86
2026-08	2	Tefal Official Store	Tefal Flagship	13,451	433.90	1.87	231.91	over 50	21,456	692.13	2.65	261.66	over 50	1.60
Columns in the file: order_month, rank_in_month, s_shop_id, shop_name, seller_type, x_shop_id, x_shop_name, days_in_month, s_gmv, s_order_count, s_adg, s_ado, s_abs, s_abs_tier, x_gmv, x_order_count, x_adg, x_ado, x_abs, x_abs_tier, gmv_ratio_x_vs_s. Full result: `results/q3_result.csv`.														

Query (`sql/04_q3_s_vs_x_performance.sql`):

```
WITH shop_month AS (                      -- same as A2
    SELECT o.order_month, o.s_shop_id, m.x_shop_id,
           SUM(o.gmv) AS s_gmv, COUNT(*) AS s_order_count
    FROM fact_s_order AS o
    JOIN fact_s_x_monthly_shop_mapping AS m
      ON m.s_shop_id = o.s_shop_id AND m.mapping_month = o.order_month
    WHERE o.order_status = 'complete'
      AND o.order_month IN ('2026-08', '2026-09', '2026-10')
    GROUP BY o.order_month, o.s_shop_id, m.x_shop_id
),
ranked AS (
    SELECT sm.*,
           EXTRACT(DAY FROM (TO_DATE(sm.order_month || '-01', 'YYYY-MM-DD')
                             + INTERVAL '1 month' - INTERVAL '1 day'))::int AS days_in_month,
           ROW_NUMBER() OVER (PARTITION BY sm.order_month ORDER BY sm.s_gmv DESC) AS rank_in_month
    FROM shop_month AS sm
),
top20 AS (SELECT * FROM ranked WHERE rank_in_month <= 20),
joined AS (                               -- new in A3: X performance for the same shop-month
    SELECT t.*, x.estimated_gmv AS x_gmv, x.estimated_order_count AS x_order_count
    FROM top20 AS t
    LEFT JOIN crawled_fact_x_shop AS x
      ON x.x_shop_id = t.x_shop_id AND x.crawled_month = t.order_month
)
SELECT j.order_month, j.rank_in_month, j.s_shop_id, s.shop_name, s.seller_type,
       j.x_shop_id, xd.shop_name AS x_shop_name, j.days_in_month,
       j.s_gmv, j.s_order_count,
       ROUND(j.s_gmv::numeric         / j.days_in_month, 2)  AS s_adg,
       ROUND(j.s_order_count::numeric / j.days_in_month, 2)  AS s_ado,
       ROUND(j.s_gmv::numeric         / j.s_order_count, 2)  AS s_abs,
       CASE WHEN j.s_gmv::numeric / j.s_order_count < 5  THEN 'under 5 USD'
            WHEN j.s_gmv::numeric / j.s_order_count < 10 THEN '5 - 10 USD'
            WHEN j.s_gmv::numeric / j.s_order_count < 50 THEN '10 - 50 USD'
            ELSE 'over 50 USD' END                          AS s_abs_tier,
       j.x_gmv, j.x_order_count,
       ROUND(j.x_gmv::numeric         / j.days_in_month, 2)  AS x_adg,
       ROUND(j.x_order_count::numeric / j.days_in_month, 2)  AS x_ado,
       ROUND(j.x_gmv::numeric / NULLIF(j.x_order_count, 0), 2) AS x_abs,
       CASE WHEN j.x_gmv IS NULL OR j.x_order_count IS NULL OR j.x_order_count = 0 THEN 'no X data'
            WHEN j.x_gmv::numeric / j.x_order_count < 5  THEN 'under 5 USD'
            WHEN j.x_gmv::numeric / j.x_order_count < 10 THEN '5 - 10 USD'
            WHEN j.x_gmv::numeric / j.x_order_count < 50 THEN '10 - 50 USD'
            ELSE 'over 50 USD' END                          AS x_abs_tier,
       ROUND(j.x_gmv::numeric / NULLIF(j.s_gmv, 0), 2)      AS gmv_ratio_x_vs_s
FROM joined AS j
JOIN dim_s_shop AS s        ON s.s_shop_id  = j.s_shop_id
LEFT JOIN crawled_dim_x_shop AS xd ON xd.x_shop_id = j.x_shop_id
ORDER BY j.order_month, j.rank_in_month;
```

Checks run: still 60 rows after the LEFT JOIN; one x_shop_id per row (mapping is 1:1 per month); the shop whose crawl is missing in October shows NULL and 'no X data'; s_abs_tier equals A2's abs_tier for the same shop-month.

A4. Detecting and reconciling crawler vs market research disagreement on X GMV

Two estimates of the same thing (X shop GMV per month): the crawler infers GMV from public "sold" counters and prices (granular, timing-sensitive, starred as primary); the vendor model is aggregated and can lag. Neither is ground truth. The goal is not "the right number" but a rule, per shop-month, for which number to use and how much to trust it.

Detection criteria, in this order

#	Criterion	Rule	Rationale
C1	Coverage	Shop-month exists in only one source → `single_source`. Not a disagreement; cannot be reconciled.	You cannot compare what you do not have. Separate first or it pollutes the gap statistics (50 of 399 crawler rows in the mock data have no MR row).
C2	Relative gap	gap = |MR − crawler| / max(MR, crawler). <10% `consistent`; 10-25% `watch`; ≥25% `conflict`.	Both sources carry ~10% error each; two errors combine to ~20%, so below 25% is noise. max() in the denominator keeps the gap symmetric whichever side is bigger.
C3	Materiality floor	Apply C2 only when max(MR, crawler) ≥ USD 500 / month; else `immaterial`.	A 60% gap on a USD 100 shop is a USD 60 problem. Keeps the review list short and actionable. Threshold is a parameter.

C4	Persistence	`conflict` in ≥2 consecutive months → `systematic`; otherwise `one_off`.	One bad month is a crawl hiccup; repeated conflict means a source is structurally wrong for that shop (wrong mapping, vendor model). Different fix, so different flag.
C5	Global bias / drift	Monthly mean and spread of (MR − crawler)/crawler across consistent shops; mean beyond ±5% or spread jumping → `source_drift`.	If every shop moves the same way, a source changed, not the shops. Detected at population level.
C6	Internal plausibility	Per source: implied ABS outside the shop's category price band; orders = 0 with GMV > 0; GMV jump > 3× vs the same source's prior month → `implausible_<source>`.	A gap says *that* they differ, not *which* is wrong. Internal checks give a verdict on each source alone.
C7	S-side triangulation (dual shops)	Compare both X numbers with the same seller's S profile (GMV, ABS, orders). The source closer to S is more credible.	Same seller, same products, same prices; S data are real transactions. Available for the ~60% of shops the business team cares about.

Reconciliation: which number to use

Never average blindly: averaging a good and a broken estimate yields a wrong number that looks reasonable. Pick one source per shop-month by rule and record the choice.

Flags	Use	Confidence	Why
consistent / watch	Crawler (keep MR as reference)	High / Medium	Agree within noise; crawler is primary and more granular
conflict + implausible_crawler	MR	Medium	Crawler row is internally broken that month
conflict + implausible_mr	Crawler	Medium	Vendor model is off for that shop
conflict, MR ≈ crawler of previous month	Crawler	Medium	Vendor lag (Appendix A: "possible lag vs real performance")
conflict, both plausible, dual shop	Source closer to S profile (C7)	Medium	Triangulate with real data
conflict, both plausible, non-dual, one_off	Crawler, flagged	Low	Default to primary; exclude from top-N views that month
conflict + systematic	**Neither; escalate**	None	Likely wrong mapping or dead listing; fix upstream, keep out of comparisons
single_source	The one available	Low	Reported as unverified

Output is one table at shop-month grain: `x_shop_id, month, crawler_gmv, mr_gmv, gap_pct, flags, gmv_final, source_used, confidence`. Downstream queries (A3, dashboards) read `gmv_final` and `confidence`, never the raw sources, so every number traces to a rule. Operating rules: run the day after both sources land; review only `conflict` and `source_drift`; calibrate thresholds on 3-6 months of history (if >5% of material shop-months are in conflict, the bar is too tight); every `systematic` case must produce an upstream fix or it returns next month. On the mock data: 339 consistent, 7 watch, 3 conflict (all 2026-09, MR = 2.2× crawler, none persisting), 50 crawler-only, mean bias +1.7%, no drift. Full note with an illustrative query: `q4_reconciliation/` and the Google Doc.

B. Projection model for company X: GMV, orders, P&L

Setup. X is assumed to be TikTok Shop Vietnam (second-largest platform, 41% of 4-platform GMV in 2025 per Metric). Quarterly model, Q1 2025A to Q4 2027E, three scenarios, USD at 26,000 VND/USD, P&L lines (1) to (13) exactly as Appendix B. Market GMV for a projected quarter = same quarter of the prior year × (1 + growth), so 2025 seasonality (Q4 ≈ 29% of the year) carries forward. X GMV = market × X share. Orders = GMV / AOV. Every input is blue, with its source or rationale beside it. The sheet has 9 tabs: README, Assumptions, Model, Scenarios, Findings, Sources, plus an extra S model and an S vs X comparison (not asked; added because "second largest" only means something relative to the largest).

Key assumptions (Base case; Worst / Best in brackets)

Input	Value	Basis
Market GMV 2025A (4 platforms)	USD 16.5bn; Q1-Q4: 3.90 / 3.88 / 3.99 / 4.76	Metric.vn quarterly reports (VND 429.7tn)
Market growth 2026 / 2027	+28% / +22% (20-33% / 15-28%)	Metric Q1 2026 forecast +32.7%; e-Conomy SEA 2026 +22%, ~17% CAGR to 2030
X share 2025A by quarter	35% / 43% / 41% / 45%	Metric; Q2 and Q4 derived from H1 and FY shares
X share 2026 / 2027	45% / 47% (42-48% / 42-52%)	Exit rate ~45%; best case = overtakes incumbent by 2027 (Q&Me "duopoly")
AOV	USD 5.5 in 2025, +3%/yr (0-5%)	Derived from Metric items sold; Mall mix rising
Commission (5), blended	3.5% 2025 → 5-7.5% through 2026 → 7.5%	VnExpress Mar 2025; Cube Asia take-rate tracker; VN Market Insights
Transaction fee (3)	4-5% 2025 → 6% from May 2026	Cube Asia; VN Market Insights
Handling fee (4)	VND 3,000 per delivered order from Q4 2025; 88% delivered	VN Market Insights
Ads (6), extra vouchers (8), other (9)	2.0% GMV; 3-4% fee on 25% enrolled GMV; 0.5%	Sea Ltd VAS ≈ 2% of GMV; Voucher Xtra terms
Voucher subsidy (11)+(12)	9% of GMV 2025 → 7.0% 2026 → 6.2% 2027 (8.5-7.8% / 6.8-5.3%)	Anchored on Sea Ltd S&M 2.8% of GMV; X in share-acquisition mode spends above

Key outputs

Fiscal year	FY2025A	FY2026E Base	FY2027E Worst	FY2027E Base	FY2027E Best
X GMV, USD bn	6.82	9.52	9.6	12.1	14.6
Orders, m	1,241	1,680	1,741	2,079	2,412
(1) Revenue, USD m	742	1,656	1,737	2,272	2,833
Take rate, % GMV	10.9%	17.4%	18.1%	18.7%	19.4%
(10) Voucher cost, USD m	610	666	747	752	775
(13) P&L, USD m	131	990	990	1,520	2,058
P&L margin, % GMV	1.9%	10.4%	10.3%	12.5%	14.1%

Findings

- **GMV:** USD 6.8bn (FY2025A) to USD 9.6-14.6bn in FY2027E; Base USD 12.1bn, 33% two-year CAGR. The worst-to-best spread (USD 5bn) is ~45% market size and ~55% share, so the share assumption matters more than the market forecast.
- **Seasonality:** Q4 is ~29-32% of the year. Q4 2027E Base = USD 3.5bn in one quarter, more than Q1+Q2 2025 combined. Annual targets that ignore this under-plan Q4 fulfilment and subsidy by about a third.
- **Orders** grow slower than GMV (1.7× vs 1.8×) because AOV rises with Mall mix. Peak ADO 6.5m orders/day in Q4 2027E vs 4.3m in Q4 2025: fulfilment capacity binds before demand.
- **Take rate** is policy-driven, not volume-driven: 8.6% (Q1 2025) to 18.8% (Q4 2026E), +10pp, from three dated fee changes. After that revenue growth is pure volume.
- **P&L** under Appendix B is contribution before fulfilment, payment, headcount and tech. X turns positive from Q2 2025 (after the April 2025 commission rise), FY2025A ~USD 130m (1.9% GMV), FY2027E Base USD 1.5bn

(12.5%). A full P&L would be several points lower (Sea Ltd adj. EBITDA 0.7% of GMV).

- **Most sensitive inputs:** subsidy (1pp of GMV = USD 121m in FY2027E; holding 2025 intensity cuts P&L to USD 1.2bn) and commission (40% of FY2027E revenue; a 2pp cut removes ~USD 240m, the size of the whole ads line).
- **Implication for S's seller acquisition:** X's take rate now equals or exceeds S's, so "lower fees on X" no longer holds. S should sell fee stability and lower subsidy dependence to the Mall and high-ABS sellers surfaced in A2/A3, whose margins are hit hardest by a 6% transaction fee, and push before Q4 when X's subsidy budget is most stretched.

Extra: S vs X on the same market and scenario (Base case)

FY	GMV S / X, USD bn	X / S	Orders S / X, m	Take rate S / X	Voucher % GMV S / X	P&L S / X, USD m
2025A	9.26 / 6.82	0.74×	1,424 / 1,241	15.0% / 10.9%	5.1% / 8.9%	918 / 131
2027E	13.16 / 12.13	0.92×	1,946 / 2,079	18.9% / 18.7%	4.5% / 6.2%	1,896 / 1,520

X closes the GMV gap but does not pass S in the Base case (only in the Best case, 1.13×, from 2027). X leads on orders from 2026 because its AOV is ~15% lower. By 2027 both charge ~19% of GMV; the remaining P&L gap is subsidy, not fees. S share is derived as 1 − X − others (Lazada, Tiki ≈ 2-3%), so it inherits any error in the X share.

Caveats and sources

Market figures are third-party estimates (Metric, Q&Me) that differ by ~3 share points for the same period; Metric is used for quarterly alignment. AOV is derived, not disclosed. Fees are blended across categories from published ranges. Subsidy ratios are assumptions anchored on Shopee's marketing-to-GMV benchmark. Treat FY2027 outputs as ranges. Sources (full URLs and access dates on the Sources tab): Metric.vn via VnEconomy (31 Jan 2026) and CafeF (29 Oct 2025); Q&Me via The Investor (3 Jun 2026); e-Conomy SEA 2026 via Vietnam News (6 Nov 2026); Cube Asia TikTok Shop take-rate tracker (Aug 2026); VN Market Insights TikTok Shop fee guide (2026); VnExpress International (17 Mar 2025); Sea Limited FY2025 results (2 Mar 2026); Eimskip Vietnam, Shopee fee schedule (20 Jun 2026).